Review Data for the article "Gender health-mortality paradox rests on false premises: results of a scoping review"

Error codes and coding rules

Every definition has two parts: a claim that women and men differ in health, and an assumption about how health and mortality are related in the two groups. E1 concerns the first, E2–E4 the second.

E1 - health is not comparable between women and men.

- **E1a**: there is no common standard for health across the sexes, because health is judged against a standard that is itself sex-specific.
- **E1b**: where health is self-reported, the same underlying state is recorded differently by women and men, through reporting thresholds, symptom attribution and differential health-care contact.
- **E1c**: where health is a composite, the conditions or deficits summed are not the same for women and men, so the totals are not two values of one measure.

E2 - the inference between health and mortality.

- **E2a**: from health to mortality. Health is assumed to predict mortality in the same way for women and men.
- **E2b**: from mortality to health. Mortality is assumed to reveal health — that a survival difference identifies a health difference — when mortality also depends on factors that differ between the sexes.
- **E2c**: an individual-level hazard relation is assumed to hold between group averages. This would require both that the relation is not modified by sex and that it is linear in the health measure.

E3 - longer life is assumed to imply a smaller share of life in poor health (compression of morbidity).
E4 - the comparison is made among people who have a given disease, a population that women and men enter differently, so the two groups compared differ in the other determinants of survival.

Assumptions: by "prevalence of morbidity" the authors mean "prevalence of chronic diseases" and code it as E1c - this rule does not extend to "disability"; "suicidal behaviour" is "prevalence of poor health".
Frailty: E1b is assigned where the definition names a frailty index or a level or burden of frailty, since these are built from self-reported conditions unless the authors specify otherwise. Where the definition says only that women are "frailer" or "more frail", no measure is named and neither E1b nor E1c is assigned.
Direction of the inference: The direction separates types 1 and 2. Where a definition contains a concessive marker ("although", "despite", "in spite of", "even though", "even if"), we took the clause it attaches to as the one given, and the other term as the contradicting observation. Where no concessive marker is present, we assigned the definition on our reading of which term the sentence presents as the one taken as given. The remaining types are identified not by direction but by the health term (type 3), the level at which the major premise is stated (type 4), the absence of a specified measure (type 5), or the population within which the comparison is made (type 6).

Type 1. From health to mortality: Poorer health is taken as given; lower mortality is the observation that contradicts it. Health is assumed to predict mortality in the same way for women and men. The health measure varies — morbidity, disability, frailty, self-rated health — and mortality is expressed as a rate or as length of life.

Citation	**Text defining the paradox**	**Errors**
Bastos et al. (2015, p.2)	The higher prevalence of morbidity in women and, on the other hand, the higher male mortality rate is a contrast that (...) has been evidenced in several health research called gender paradox.	E1a, E1c, E2a
Belon et al. (2014, p.2)	This contrast between female disadvantage in morbidity and female advantage in mortality is known as the male-female health-survival paradox.	E1a,E2a
Berardelli et al. (2022, p.1)	Gender paradox in suicidology, an important sex difference has been reported for suicidal behavior. Females have higher rates of suicide ideation and suicide attempts than males, while fatal suicide acts are typically higher for males than for females.	E1a, E2a
Biz et al. (2024, p.1)	The review addresses the morbidity-mortality paradox, where women, despite appearing less healthy on frailty indexes, show lower mortality rates.	E1a, E1b, E2a
Biz et al. (2024, p.8)	Despite scoring higher on the frailty index, suggesting poorer health, including poorer bone and cartilage health, women experience lower mortality rates across all age groups, a phenomenon referred to as the "morbidity-mortality paradox".	E1a, E1b, E2a
Canetto and Sakinofsky (1998, p.1)	Puzzling paradox in the epidemiology of suicidal behavior. In most countries where the prevalence of suicidality has been studied, females have higher rates of suicidal ideation and behavior than males, yet mortality from suicide is typically lower for females than for males.	E1a, E2a
Chen et al. (2022, p.22)	The phenomenon of the "male-female health-survival paradox", which states that females experience higher rates of health conditions but live longer than males.	E1a, E1c, E2a
Fridh et al. (2022, p.2)	The fact that psychological distress is more often reported by women than men while mortality rates are higher among men is an example of the male-female health-survival paradox.	E1a, E1b, E2a
García-Calvente et al. (2012, p.913)	In Spain, there is also the paradox that, although women show worse perceived health, mortality rates are higher for men.	E1a, E1b, E2a
Gordon and Hubbard (2020, p.183)	Compared with age-matched men, women tend to have poorer health status (ie, they are more frail) but longer life expectancy (ie, they are more resilient).	E1a, E2a

Citation	**Text defining the paradox**	**Errors**
Green and Pope (1999, p.1363)	Women use more medical services and report more morbidities than do men and, paradoxically, that they live longer.	E1a, E1b, E1c, E2a
Kulminski et al. (2008, p.1052)	The traditional sex morbidity-mortality paradox that females have worse health but better survival than males.	E1a, E2a
Lakomý and Petrová Kafková (2017, p.373)	Older women are frailer, have poorer health and functional status (...) Despite all this adversity, women have lower mortality rates than men in all age categories and across all causes of death. This phenomenon is often called the "gender paradox"	E1a, E2a
Li et al. (2024, p.2)	This finding presents a paradox: A higher level of physiological dysregulation in females appears to contradict their longer life expectancy.	E1a,E1c, E2a
Luy and Minagawa (2014, p.12)	The literature suggests that women report worse health but live longer than men - a phenomenon known as the gender paradox in health and mortality.	E1a, E1b, E2a
Oksuzyan et al. (2008, p.91)	There is a remarkable discrepancy between the health and survival of the sexes: men are physically stronger and have fewer disabilities, but have substantially higher mortality at all ages compared with women: the so-called male-female health-survival paradox.	E1a, E2a
Oksuzyan et al. (2009, p.504)	Apparent contradiction "men report better health than females, but women still outlive men in all countries around the world" (...) the health-survival paradox	E1a, E1b, E2a
Oksuzyan et al. (2014, p.243)	In high income countries females outlive men, although they generally report worse health, the so-called male-female health-survival paradox.	E1a, E1b, E2a
Perelman et al. (2012, p.2347)	the paradox is still why men show shorter life expectancy despite their better health.	E1a, E2a
Phillips et al. (2023, p.1)	Despite consistently reporting poorer health, women universally outlive men. (...) might explain this paradox.	E1a, E1b, E2a
Phillips et al. (2023, p.2)	The paradox of women's poorer subjective health but greater longevity.	E1a, E1b, E2a
Scheel-Hincke et al. (2021, p.1281)	The contradictory phenomenon that women report poorer health but live longer than men has been denoted the male-female health-survival paradox.	E1a, E1b, E2a
Singh-Manoux et al. (2008, p.2251)	This discrepancy between morbidity and mortality rates is referred to as the "gender paradox", which refers to higher morbidity but lower mortality among women than among men.	E1a, E1c, E2a
Theou et al. (2015, p.476)	The phenomenon that women have poorer health but longer life expectancy, known as the male-female health survival paradox.	E1a, E2a

Citation	Text defining the paradox	Errors
Zeng et al. (2024, p.2)	The phenomenon that females on average live significantly longer than males in spite of having poorer health and lower socioeconomic status is known as the male-female health-survival paradox.	E1a, E2a

Type 2. From mortality to health: Lower mortality is taken as given; poorer health is the observation that contradicts it. Mortality is assumed to reveal health. Measures vary as in type 1; what separates the two types is the direction of the inference, and with it which assumption the argument requires.

Citation	Text defining the paradox	Errors
Adjei et al. (2017, p.11)	The health paradox in contemporary welfare countries [47], that women live longer yet they report poorer health.	E1a, E1b, E2b
Ahrenfeldt et al. (2019, p.1025)	Despite the longer life expectancy, women generally report worse health than men "the so-called male-female health survival paradox."	E1a, E1b, E2b
Alberts et al. (2014, p.339)	The male-female health-survival paradox - the phenomenon observed in modern human societies in which women experience greater longevity and yet higher rates of disability and poor health than men.	E1a, E2b
Arber and Cooper (1999, p.61)	The well known paradox that men are more likely to die than women, but that women suffer from higher levels of morbidity than men.	E1a, E1c, E2b
Archer et al. (2018, p.4)	Human females tend to live longer but are in poorer health than males late-in-life.	E1a, E2b
Arosio et al. (2020, p.1)	The "male-female health-survival paradox" evidences that the survival advantage observed in women is linked to higher rates of disability and poor health status compared to men.	E1a, E2b
Arosio et al. (2020, p.5)	The sex-frailty paradox is a well-described phenomenon, in which women have experienced greater longevity than men since the 18th century. However, this survival advantage in females is linked to higher rates of disability and poor health status during their lives, as demonstrated by the higher FI shown by women compared men in this cohort.	E1a, E2b
Austad (2006, p.79)	Paradoxically, although women have lower mortality rates they have higher overall rates of physical illness than do men.	E1a, E1c, E2b
Austad and Fischer (2016, p.1027)	For all their robustness relative to men in terms of survival, women on average appear to be in poorer health than men throughout adult life.	E1a, E2b

Citation	Text defining the paradox	Errors
Austad and Fischer (2016, p.1022)	Paradoxically, although women live longer, they suffer greater morbidity particularly late in life.	E1a, E1c, E2b
Avdic et al. (2019, p.2)	The morbidity-mortality paradox (...) women outlive men, while having worse morbidity outcomes.	E1a, E1c, E2b
Balard et al. (2011, p.1)	Gender paradox in old age. Even if women are [more numerous in old age and] live longer than men, men are in better physical and cognitive health, are higher functioning, and have superior vision.	E1a, E2b
Bambra et al. (2021, p.17)	"gender health paradox" (...) men have shorter life expectancies and higher mortality rates than women, and yet, women report higher morbidity.	E1a, E1b, E1c, E2b
Chun et al. (2006, p.576)	Despite women's lower mortality, they seemed to have higher rates of morbidity than men.	E1a, E1c, E2b
Chung et al. (2023, p.22)	Male-female health-survival paradox that women in Hong Kong are living a longer life but with greater disability than men.	E1a, E2b
DeWitte (2024, p.1)	In many human populations today, females experience lower age-specific mortality rates compared to males at most, if not all ages, and women live longer on average than men (...). However, in some of those same populations, women tend to experience higher burdens of disease and disability and poorer self-rated health.	E1a, E1b, E2b
Draper and Abrahamson (2014, p.655)	The paradoxical observation that in old age there are more women than men (= women have lower mortality rates) but the men are in better health.	E1a, E2b
Erving (2011, p.384)	Women's lower mortality rate but higher morbidity rate, compared to men, is referred to as the gender-health paradox	E1a, E1c, E2b
Gordon et al. (2017, p.31)	Despite "health advantages" that contribute to longer life expectancy, females have greater levels of disability, more co-morbidities and poorer self-rated health.(...) the "male-female health-survival paradox."	E1a, E1b, E1c, E2b
Gorman and Read (2006, p.96)	Women report worse health than men despite the fact that they live longer (Verbrugge 1985), a phenomenon known as the "morbidity paradox."	E1a, E1b, E2b
Gu et al. (2009, p.2171)	The gender paradox in health is widely recognized. However, the reasons why women enjoy greater longevity but worse health are complex.	E1a, E2b

Citation	Text defining the paradox	Errors
Hammond and Renzi-Hammond (2023, p.720)	The mortality-morbidity paradox refers to the inconsistency in survival and disease between males and females: females live longer but tend to suffer greater age-related disease and disability.	E1a, E1c, E2b
Hoogendijk et al. (2019, p.80)	This is the phenomenon that higher life expectancy is accompanied by higher rates of poor health in women compared to men. This is seen as a paradox, since poor health is usually associated with lower survival.	E1a, E2b
Hubbard and Rockwood (2011, p.204)	Although women live longer than men, they tend to have poorer health status. This well described phenomenon has been termed the male-female health-survival paradox.	E1a, E2b
Kane and Howlett (2021, p.1)	Clinical studies have used these tools to show that women are frailer than men even though they have longer lifespans. Still, factors responsible for this frailty-mortality paradox.	E1a, E2b
Kaplan et al. (1991, p.86)	Although women live longer, it has been reported that they paradoxically experience more physical and psychological illnesses.	E1a, E1c, E2b
Karun et al. (2025, p.2)	This phenomenon of female living longer than male (mortality advantage) but simultaneously being more prone to having worse health (disability disadvantage) than male is widely known by several names, including the "male-female health-survival paradox" (...).	E1a, E2b
Keevil et al. (2013, p.5)	Male-female health-survival paradox, since women are known to have longer life expectancies than men but lower physical capability (...)	E1a, E2b
Mayor (2015, p.1)	Women have a life-expectancy advantage over men, but a marked disadvantage with regards to morbidity. This is known as the female-male health-survival paradox.	E1a, E1c, E2b
Newman and Brach (2001, p.345)	In spite of the fact that women seem to be hardier and to have a better survival rate than do men, the prevalence of disability is greater in older women. (...) may explain this paradox.	E1a, E2b
Nusselder et al. (2019, p.917)	Women experience lower mortality, but report more disability compared to men of the same age. (...) the "health-survival paradox".	E1a, E1b, E2b

Citation	Text defining the paradox	Errors
Okamoto et al. (2021, p.1)	Females generally live longer than males, but women tend to suffer from more illnesses and limitations than men do, also for dementia.	E1a, E1c, E2b
Oksuzyan et al. (2010, p.213)	Men have higher death rates than women, but women do worse with regard to physical strength, disability, and other health outcomes, the so called male-female health-survival paradox.	E1a, E2b
Oksuzyan et al. (2015, p.1)	The apparent contradiction that women live longer but have worse health than men, the so called male-female health-survival paradox.	E1a, E2b
Olsen et al. (2023, p.35)	(...) male-female health survival paradox-the fact that women outlive men in almost all countries in the world, but at the same time tend to report poorer health.	E1a, E1b, E2b
Ostan et al. (2016, p.1714-1715)	Women pay for their survival advantage (...), women reported a poorer self-perceived health, more long-standing illnesses and/or more health problems than men [10]. A possible explanation for this gender associated health-survival paradox (...).	E1a, E1b, E1c, E2b
Padua et al. (2018, p.510)	We have explored, the male-female health-survival paradox, observed in several reports (...), which defines the tendency of the higher mortality, but lower disability in men as compared with women.	E1a, E2b
Pongiglione et al. (2016, p.736)	The gender paradox in health and mortality (...) reflects the finding that women live longer than men, but tend to have more disability than males.	E1a, E2b
Redondo-Sendino et al. (2006, p.2)	Compared to men, women live longer but, paradoxically, report greater morbidity and disability and make greater use of health-care services at the end of life	E1a, E1b, E1c, E2b
Rieker and Bird (2005, p.40)	Women live longer than men, yet they have higher morbidity rates and in later years a diminished quality of life. The paradox of men's higher mortality and lower morbidity compared with women.	E1a, E1c, E2b
Scheel-Hincke et al. (2020, p.69)	The phenomenon that women outlive men, but at the same time suffer from more disabilities and are physically weaker, is denoted the male-female health survival paradox.	E1a, E2b
Stephan et al. (2021, p.1)	The male-female health-survival paradox: Although on average women live longer than men (...), women report higher age-specific morbidity burden.	E1a, E1b, E1c, E2b

Citation	Text defining the paradox	Errors
Takahashi et al. (2020, p.1)	The "gender paradox" whereby women enjoy longer life expectancy, but worse self-rated health compared to men.	E1a, E1b, E2b
Van Oyen et al. (2013, p.143)	In many countries the mortality advantage of women is balanced by a disability disadvantage. This contrast is called the female-male health-survival paradox	E1a, E2b
Vidal-Meliá et al. (2024, p.2814)	"Male-female health-survival paradox", a phenomenon seen in developed countries where women have greater longevity but higher rates of disability and poorer health than men.	E1a, E2b
Walter-Ginzburg et al. (2005, p.1705)	Aging is characterized by a "gender paradox" with higher life expectancy yet greater morbidity for women as compared to men.	E1a, E1c, E2b
Yong et al. (2011, p.189)	Gender health-survival paradox found in Western countries - that women have higher morbidity rates despite longer life expectancy.	E1a, E1c, E2b

Type 3. From life expectancy to health expectancy: Longer life is taken as given; more years lived in poor health is the observation. Health expectancy is computed from a life table and so is not independent of mortality. These definitions assume that longer life brings a smaller share of life in poor health.

Citation	Text defining the paradox	Errors
Bambra et al. (2021, p.18)	Men have lower life expectancies than women, but women spend their extra years with higher levels of ill health: the so-called "gender health paradox".	E1a, E3
Cambois et al. (2019, p.839)	the male-female health-survival paradox applies in this rural population; women live longer than men, but they also live longer with disability.	E1a, E3
Chen et al. (2025, p.2)	Despite men's higher mortality rates, women are more likely to experience worse health, spending a larger proportion of their remaining life in poor health. (...) gender paradox.	E1a, E3
di Lego et al. (2021, p.1)	The male-female health-mortality paradox results from the fact that females live longer than males but spend a higher proportion of their total life expectancy in poorer health states.	E1a, E3
Kingston et al. (2014, p.1)	Women live longer than men on average, but their longer life expectancy is accompanied by more years with disability, both in absolute terms and as a proportion of remaining life. Understanding the basis of this "disability-survival paradox".	E1a, E3
Merrick and Brayne (2024, p.S3)	The "male-female health-survival paradox" is demonstrated whereby women live longer but with more comorbidity and disability.	E1a, E1c, E3

Citation	Text defining the paradox	Errors
Shoko (2018, p.324)	"health-survival paradox" (...), whereby females live longer than males on average but their longer life expectancy is accompanied by more years with disability, both in absolute terms and as a proportion of remaining life.	E1a, E3
Solé-Auró et al. (2022, p.1)	The gender paradox (women living longer but unhealthier)	E1a, E3

Type 4. From individual risk to group averages: The major premise is stated at the individual level while the observation compares groups, so an individual hazard relation is assumed to hold between group averages. What identifies these definitions the level at which the major premise is stated: it is a risk or vulnerability attaching to a person, or a comparison made within levels of health, rather than a rate compared between groups.

Citation	Text defining the paradox	Errors
Christensen (2008, p.1575)	The male-female health-survival paradox (i.e., males report better health than females, but encounter higher mortality at all ages)	E1a, E1b, E2c
Cohen et al. (2018, p.182)	The male-female health-survival paradox, women are more frail but despite this men are more susceptible to mortality.	E1a, E2c
Gordon and Hubbard (2018, p.10)	(...)"sex-frailty paradox": females may be considered to be more frail (because they have poorer health status) but also less frail (because they are less vulnerable to death) than males of the same age. *[Identical wording to Gordon and Hubbard (2019).]*	E1a, E2c
Gordon et al. (2018, p.13)	Even though females experience higher levels of frailty, they face a lower risk of mortality than males. This is the "male-female health-survival paradox".	E1a, E1b, E2c
Gordon and Hubbard (2019, p.44)	(...)"sex-frailty paradox": females may be considered to be more frail (because they have poorer health status) but also less frail (because they are less vulnerable to death) than males of the same age.	E1a, E2c
Gordon and Hubbard (2020, p.184)*	In community dwelling populations aged over 65 years, women are more likely to be frail and to have a greater burden of frailty than men of the same age. Yet women appear to be more resilient at any given age or level of frailty, their mortality rates are lower. This sex-frailty paradox (...)	E1a, E1b, E2c
Veronese et al. (2019, p.82)	These findings overall suggest that women have a poorer health status and higher levels of frailty, and a lower risk of mortality, than men [2]. This phenomenon has been defined as the "male-female health-survival paradox".	E1a, E1b, E2c

*Gordon and Hubbard (2020, p.184), can be read both ways. It opens with a group comparison of frailty prevalence and closes with lower mortality rates, which is the type 1, but between the two it locates the paradox in women being more resilient at any given age or level of frailty, which is a within-stratum comparison. We assign it to type 4 on that clause. The same article states the paradox again at p.183 in a form with no stratification, and that definition is assigned to type 1.

Type 5. Under-specified terms: The paradox is stated in a fixed formula rather than in terms the author defines.

Citation	Text defining the paradox	Errors
Chen et al. (2025, p.1)	The male-female health-survival paradox is characterized by the phenomenon where "women get sicker, but men die quicker."	E1a, E2a or E2c
Luy (2024, p.291)	not own words, but citation "women get sicker, but men die quicker" from Lorber and Moore (2002)	E1a, E2a or E2c
Schunemann et al. (2017, p.80)	The inverse association between health and longevity across sexes is known as the morbidity-mortality paradox.	E1a, E2a or E2c

Type 6. Comparison within a diseased population: Women and men are compared among those who have a given disease or intervention, not in the general population.

Citation	Text defining the paradox	Errors
Boccuzzi et al. (2005, p.965)	This study supports the concept of a "gender paradox" even in a diabetic population after PCI. Women with diabetes undergoing PCI present with overall worse clinical and angiographic characteristics and may still pay a price in initial risk even with current techniques, but fare surprisingly well long-term after coronary stenting.	E1a, E4
Dekleva et al. (2024, p.1)	(The opposite the standard health-mortality paradox:) There is therefore a "sex paradox" despite the lower prevalence of obstructive CAD and HF with reduced ejection fraction (HFrEF), women have a higher mortality rate after MI.	E1a, E4
DesJardin et al. (2024, p.902)	Female sex is one of the strongest known risk factors for pulmonary arterial hypertension (PAH), and yet males with PAH have worse survival - a phenomenon which has been referred to as the "sex paradox" or "sex puzzle" in PAH.	E1a, E4
Foderaro and Ventetuolo (2016, p.1)	Female sex has been shown to be a risk factor for the development of PAH, but women have improved survival compared to men with PAH. These paradoxical observations (...)	E1a, E4
Stehli et al. (2023, p.1)	The apparent sex paradox seen in women undergoing transcatheter aortic valve implantation refers to the phenomenon of women experiencing higher rates of short-term mortality and bleeding events, but demonstrating improved long-term survival as compared to men.	E1a, E4

Paradox not stated in the article

Discusses differences in health and mortality of women and men without stating the contradiction. Not coded, as no definition of the paradox is given.

Citation	Text
Avdic and Johansson (2017, p.440)	(...) seemingly contradictory public health patterns between men and women, often called the morbidity-mortality paradox.
Dix (2014, p.213)	The abundance of data demonstrating that females are better suited to survival than males argues against any "health-survival paradox". Women outlive men because they are healthier.
Doblhammer and Hoffmann (2010, p.482)	Why women live longer than men, but suffer from worse health, is, however, still an unsolved puzzle.
Howlett et al. (2014, p.7)	Women had lower mortality than did men for any level of either FI [frailty index].
Macintyre et al. (1996, p.617)	(...) men die earlier than women, but that women have poorer health than men.
Sen et al. (2002, p.107)	(...) men die of their illnesses while women have to live with theirs.
Verbrugge (1985, p.163)	There is no contradiction between the health and mortality statistics since both point to more serious health problems for men.
Verbrugge (1989, p.282)	(...) higher morbidity and health services use for women, while mortality rates are higher for men.

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